

List of advanced mathematics courses

Course	When take	Grade	Textbook	Author	Topics/chapters covered
Probability	2020 Fall	96/100	Foundations of Probability (Chinese title: 概率论基础)	Xianping Li (李贤平)	events and probability; probability spaces; conditional probability and independence; random variables and distribution functions; moments and characteristic functions; limit theorems
Real Analysis	2021 Spring	90/100	A Concise Course in Real Analysis (Chinese title: 实变函数简明教程)	Donggao Deng (邓东皋)	Lebesgue measure; measurable functions; Lebesgue integral; Fubini's theorem; L^p space
Complex Analysis	2021 Spring	91/100	Complex Analysis (Chinese title: 复变函数)	Jiarong Yu (余家荣)	holomorphic functions; contour integrals; power series and Laurent series; residues; conformal mappings; analytic continuation
Algebra	2021 Fall	89/100	Abstract Algebra (Chinese title: 抽象代数)	Chunlai Zhao (赵春来)	groups (special classes, Sylow theorems, composition series, free groups, rotation groups); rings (ideals, factorization in integral domains); fields (field extensions, splitting fields, basics of Galois theory); modules
Partial Differential Equations	2021 Fall	99/100	Equations of Mathematical Physics (Chinese title: 数学物理方法)	Chaohao Gu et al. (谷超豪等)	wave equation; heat equation; Laplace equation; classification of second-order linear PDEs; first-order systems
Functional Analysis I	2022 Spring	90/100	Lectures on Functional Analysis (Chinese title: 泛函分析讲义)	Gongqing Zhang (张恭庆)	metric spaces; linear operators and linear functionals; compact operators
Differential	2022 Spring	96/100	Differential Geometry of	Manfredo P. do	curves and regular surfaces; Gauss map; Gauss–Bonnet theorem;

Geometry			Curves & Surfaces	Carmo	Hopf–Rinow theorem; Bonnet’s theorem
Operations Research	2022 Spring	100/100	Operations Research: A General Introduction (Chinese title: 运筹学通论)	Quanling Wei et al. (魏权龄等)	linear programming; nonlinear programming; integer programming; dynamic programming; graphs and networks
Functional Analysis II	2023 Fall	84/100	Lectures on Functional Analysis (Chinese title: 泛函分析讲义)	Gongqing Zhang (张恭庆)	distributions and distribution spaces; Banach algebras; unbounded operators
Riemannian Geometry	2023 Fall	80/100	Introduction to Riemannian Geometry (Chinese title: 黎曼几何初步)	Hongxi Wu (伍鸿熙)	exponential map; Gauss lemma; first and second variation formulas; Bonnet–Myers theorem; Morse index theorem; minimal submanifolds
Commutative Algebra	2024 Spring	96/100	Introduction to Commutative Algebra	M. F. Atiyah; I. G. Macdonald	modules; primary decomposition; Noetherian and Artinian rings; discrete valuation rings and Dedekind domains; completion (Chapter 1-10)
Modern Partial Differential Equations	2024 Spring	87/100	Lectures on Non-Linear Wave Equations	Christopher D. Sogge	energy inequalities; quasilinear equations; semilinear equations (Chapter 1-2)
Algebraic Topology	2024 Fall	99/100	Differential Forms in Algebraic Topology	Raoul Bott; Loring W. Tu	de Rham complex; Poincaré lemma; Mayer – Vietoris argument; Thom isomorphism; de Rham cohomology; spectral sequences; characteristic classes